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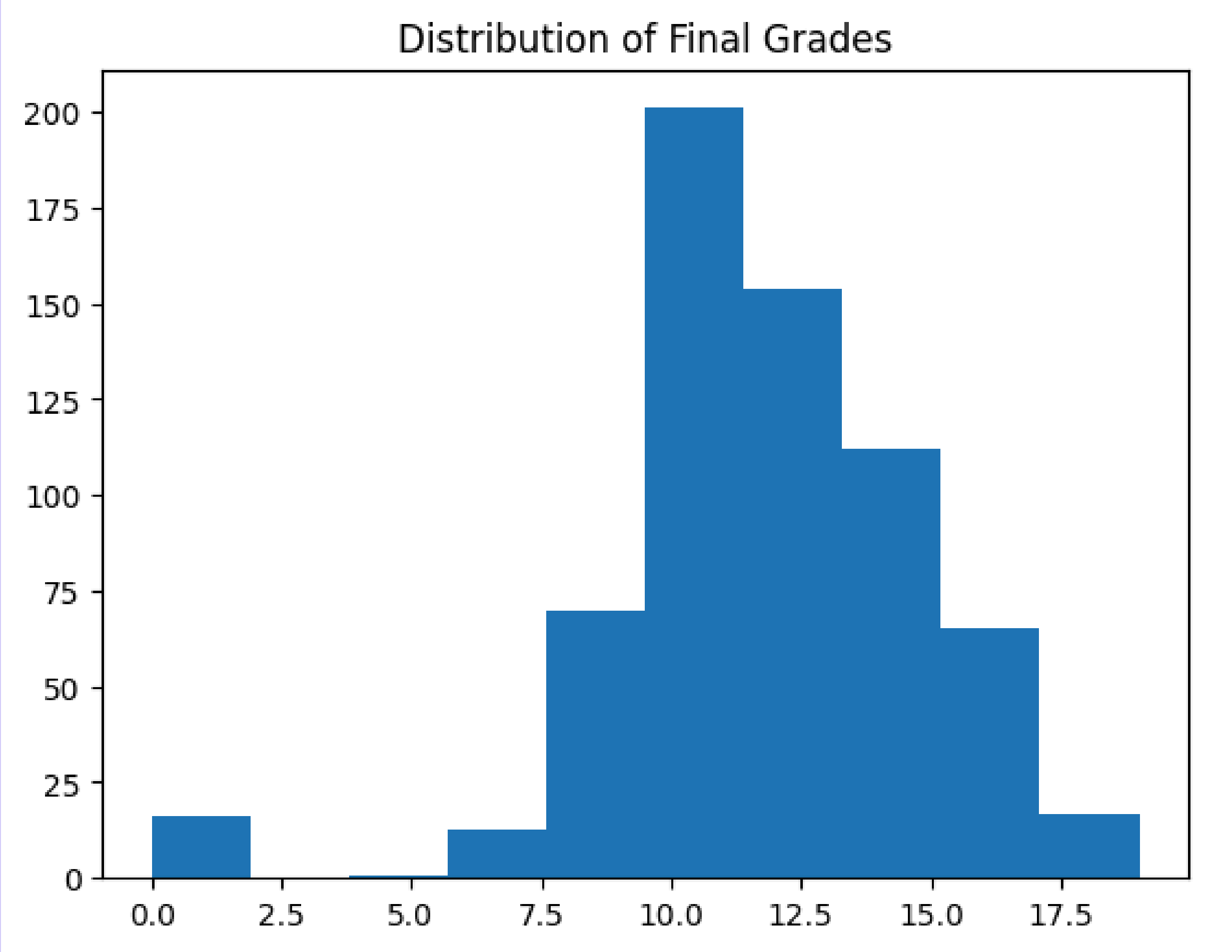
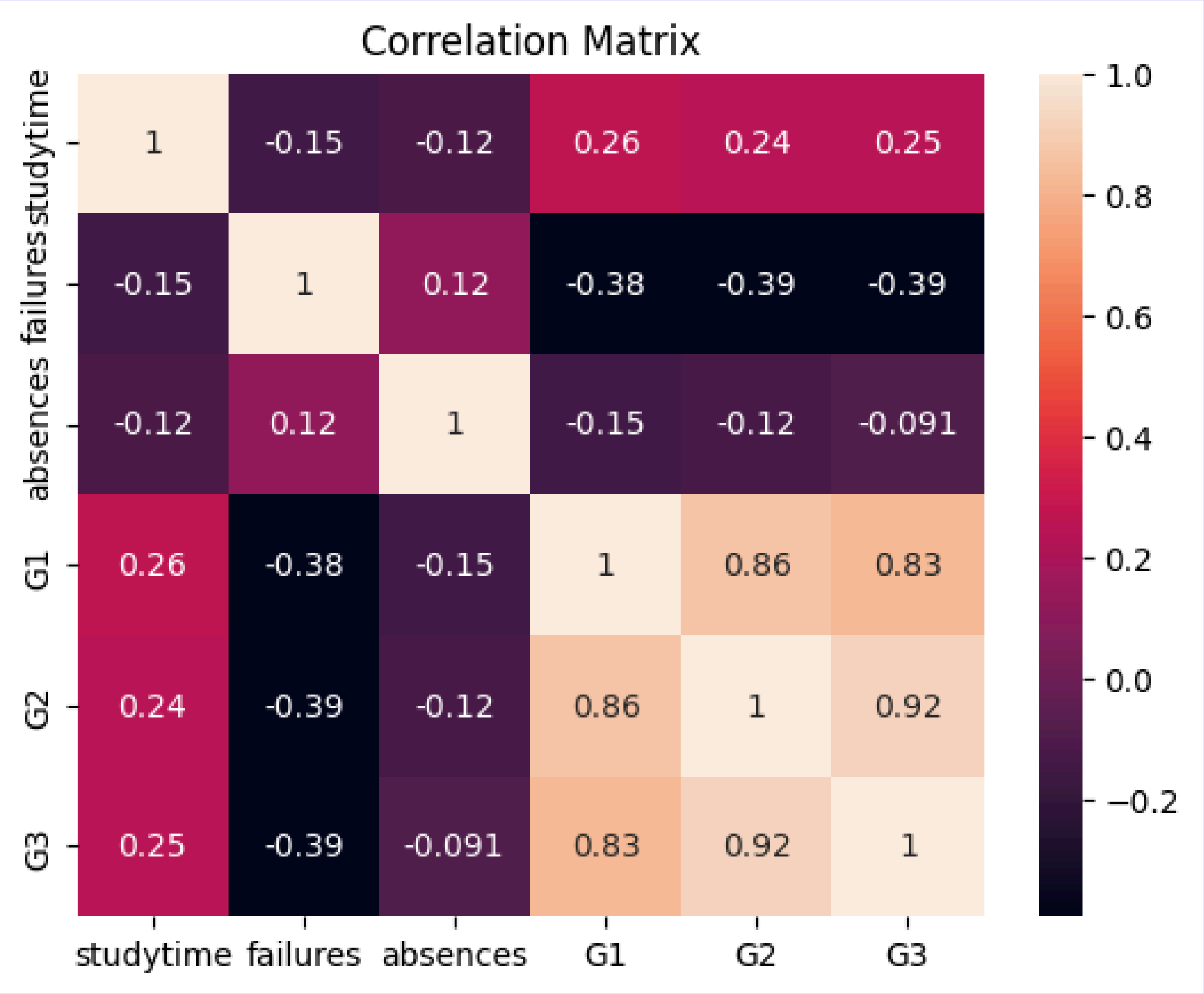
Data Bootcamp Spring ‘26

Data Bootcamp Final Project

Overview

- **The goal of this model is to predict students final grades using numeric academic variables.**
- **The model uses UC Irvings student performance dataset.**
- **Uses statsmodels Ordinary Least Squares (OLS) to perform a linear regression.**





Snapshot of the Data

- **Most students score in the mid range**
- **G1 (first period/semester grades) and G2(second period/semester grades) are strongly correlated with G3 (final grades)**

Model Results

OLS Regression Results

Dep. Variable:	G3	**R-squared:**	0.849
Model:	OLS	**Adj. R-squared:**	0.847
Method:	Least Squares	**F-statistic:**	538.8
Date:	Thu, 14 May 2026	**Prob (F-statistic):**	2.83e-194
Time:	00:43:27	**Log-Likelihood:**	-786.28
No. Observations:	486	**AIC:**	1585.
Df Residuals:	480	**BIC:**	1610.
Df Model:	5		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P>|t|**	**[0.025**	**0.975]**
const	0.0654	0.293	0.223	0.823	-0.510	0.641
studytime	0.0965	0.071	1.356	0.176	-0.043	0.236
failures	-0.2523	0.102	-2.473	0.014	-0.453	-0.052
absences	0.0203	0.012	1.699	0.090	-0.003	0.044
G1	0.1386	0.043	3.250	0.001	0.055	0.222
G2	0.8688	0.041	21.420	0.000	0.789	0.949

Omnibus:	334.986	**Durbin-Watson:**	2.002
Prob(Omnibus):	0.000	**Jarque-Bera (JB):**	7239.198
Skew:	-2.656	**Prob(JB):**	0.00
Kurtosis:	21.146	**Cond. No.**	92.1

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Conclusion

01.

The model is effective at predicting final grades using numeric academic variables given the r-squared is about .85.

02.

Previous academic performance strongly predicts final grades based on strong correlation and positive coefficients.

03.

Additional variables could improve the models prediction of final grades.